AIML4OS

Artificial Intelligence and Machine Learning for Official Statistics

Newsletter 7

Welcome to our Newsletter #7 for ESSnet AIML4OS! Here you will find project updates highlighting progress, achievements, events and all news.

PROJECT OVERVIEW

The main objectives of AIML4OS are to explore the use of Artificial Intelligence/Machine Learning (AI/ML) for the production of official statistics and to implement innovative solutions for statistical products and processes. This four-year project started in April 2024, with activities structured in the following work packages

OVERARCHING WORK PACKAGES

WP1
Project management and coordination

WP2
Communication and community engagement

WP3
ESS AI/ML lab: Technical infrastructure and organisational setup

WP4
AI/ML state-of-play and ecosystem monitoring

WP5
Standards, methodological and implementation frameworks

WP6
Knowledge repository and training material

USE CASES

WP7
AI/ML on earth observation data, satellite imagery

WP8
Statistically valid and efficient editing and imputation in official statistics by AI/ML – with a special focus on editing

WP9
Imputation focus - Statistically valid and efficient editing and imputation in official statistics by AI/ML – with a special focus on imputation

WP10
From text to code - Experiences and potential of the use of AI/ML for classifying and coding

WP11
Applying ML for estimating firm-level supply chain networks

WP12
Large language models

WP13
Generation of synthetic data in official statistics: techniques and applications

In this issue: Progress and Future Work after Three Years, First AIML4OS Funathon, Exploring GenAI, From Principal to Practical Implementation

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LOOKING AHEAD: REFLECTING ON THREE YEARS OF COLLABORATION AND SHAPING THE FUTURE

A one-day meeting in Šibenik

On 2 June, project partners met in Šibenik, Croatia, for a one-day meeting to reflect on the progress made so far and discuss the project's future. The meeting was structured around **four key questions**, covering the project outputs, standards and methodology, sustainability, and possible directions beyond the current three-year period.

Are We on the Right Track? Project Outputs

The first discussion focused on whether the project is delivering what was originally envisaged. Overall, the work was considered to be on track, while participants highlighted the need to **strengthen the links between activities and deliverables** across Work Packages.
The discussion also focused on who should benefit from the project's results, identifying NSIs, students and the project community itself as key target groups. Participants stressed the importance of **producing citable publications**, to contribute to the growing knowledge base on AI. The creation of training and example datasets for different tasks was also identified as a valuable way to make the project's results more practical and reusable.

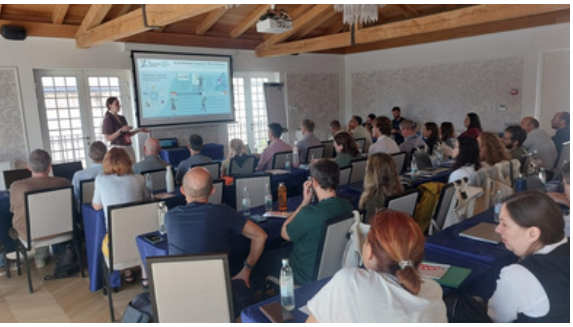
From Theory to Practice: Standards, Methodology and Implementation

The second question explored how standards and methodological work connect with the use cases developed across the project. Participants discussed the effectiveness of the TMLE model and, more broadly, the challenge of translating theoretical principles into real-world applications. Two areas emerged as particularly important for further work: **reproducibility** and the measurement of AI **maturity**. Addressing these challenges will require a collaborative approach, bringing together technical experts and combining knowledge from data science, statistical analysis and regulatory frameworks. This multidisciplinary perspective will be essential to ensure that methodological developments can be effectively implemented in official statistics.

Beyond the Project: Ensuring Sustainability

Looking beyond the project's lifetime, participants discussed how its achievements can be maintained, shared and further developed. A key question was who will take ownership of the project's repositories and technical assets once the project ends.

The project's achievements were seen as stepping stones for future developments, but their long-term use may depend on infrastructure and technologies such as Onyxia. **Establishing a community group to support continuity** was therefore proposed as one possible way to maintain collaboration and knowledge sharing. **Regional conferences** were also suggested as a means of reaching domain specialists and new communities and broadening the project's impact.







What Comes Next? A Vision Beyond Three Years

The final discussion looked further ahead, asking what could be developed once the current project comes to an end. Several emerging areas were identified as potential priorities, including **agentic AI, LLMs in production, digital sovereignty and data dissemination**.

Future work could also build on a **stronger co-development approach**, involving not only other NSIs but potentially central banks and organisations such as the JRC. Participants discussed the **possibility of benchmarking models**, countries and datasets according to the specific needs of NSIs, as well as testing the project's approaches with new NSIs.

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THE FIRST EUROPEAN AIML4OS FUNATHON

Building AI Skills Across the European Statistical System

The first European AI4OS Funathon, held on 27–28 May 2026, marked an important step in strengthening Artificial Intelligence skills across the European Statistical System. The event brought together **215 participants from 25 countries**, forming **67 international teams**. Rather than focusing on competition, the Funathon was designed as a collaborative learning initiative. Participants had the opportunity to experiment with modern AI methods, exchange knowledge with colleagues from different NSIs, and gain hands-on experience by addressing realistic statistical challenges.
Three practical projects were offered:

- **Predicting Housing Prices**: applying machine-learning algorithms to synthetic data.
- **Automatic Coding for NACE Classification**: free-text descriptions to economic activities;
- **Satellite Image Segmentation**: real satellite imagery using computer-vision methods.

High-Quality Learning Resources

Teams brought together participants with diverse professional backgrounds and varying levels of AI experience, creating an environment that encouraged peer learning, problem-solving and knowledge exchange. The quality of the educational materials was a key factor in the event's success. Participants praised the well-structured notebooks, clear explanations, practical examples and visual learning resources.
Many noted that the materials would remain valuable beyond the event itself.
The projects were designed to guide participants through complete AI workflows.



Participant Feedback

The post-event survey confirmed the positive reception of the initiative. Participants particularly appreciated:

- the opportunity to gain practical AI and machine-learning experience;
- collaboration with colleagues from across Europe;
- the quality and clarity of the learning materials;
- the availability and responsiveness of the support team;
- the friendly, inclusive and non-competitive atmosphere, which encouraged experimentation and learning.

Many participants described the Funathon as an inspiring introduction to AI applications in official statistics and valued the opportunity to learn through realistic use cases.
The positive response provides a strong foundation for future editions. Lessons learned from this first event will help enhance the participant experience, improve accessibility for newcomers and expand opportunities for advanced users.

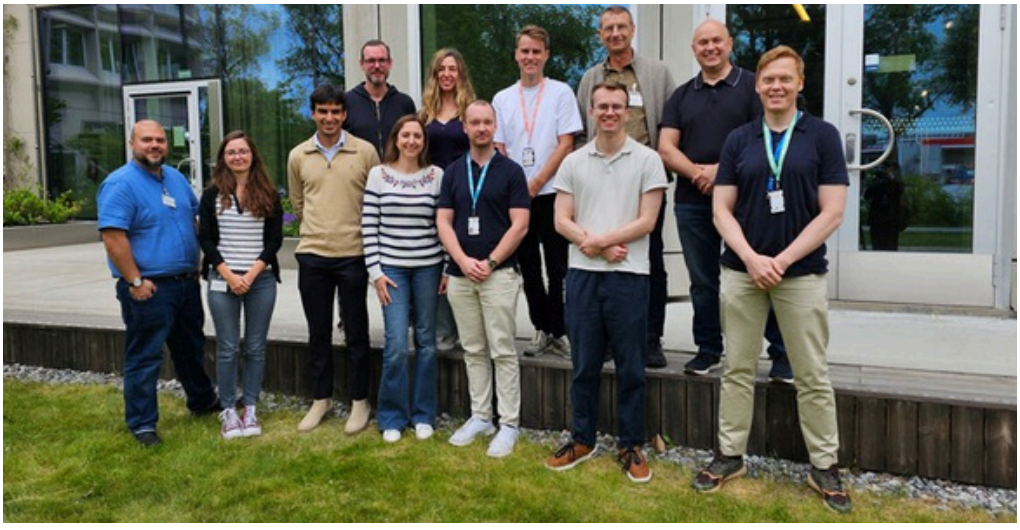
Further information, project materials and learning resources are available on the **AIML4OS Funathon website**



STOCKHOLM SPRINT ON GENERATIVE AI

Exploring agentic web scraping, AI-supported metadata management and statistical reporting assessment in NSI environments

The LLM team held a three-day sprint on Generative AI in Stockholm between 16-18 June. The sprint brought together participants from Sweden, Portugal, Slovenia, Spain, the Netherlands, France and Eurostat, with 13 participants taking part in the active development work on site. There was also interest from remote participants to listen in and follow parts of the work.
The sprint focused on three use-case areas: **agentic web scraping, AI-supported metadata management, and the use of generative AI** to assess whether media reporting on statistics is aligned with the underlying statistical releases and analysis. These use cases were selected because they represent types of tasks where traditional rule-based systems are often difficult to maintain, or where generative **AI may provide new ways to work with complex**, unstructured and context-dependent information.
Another important theme was **digital sovereignty** and **practical implementation** in environments that are realistic for national statistical institutes. The work therefore focused on approaches that could reasonably be run in an NSI-controlled infrastructure, or in an environment similar to SSPCloud.



The sprint also contributed material for upcoming deliverables, including **reflections on architecture, evaluation, documentation and practical implementation**. The results will be further documented in the repository and used as input to the continued AIML4OS work on the responsible and practical use of generative AI in official statistics.

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FROM PRINCIPLES TO PRACTICAL IMPLEMENTATION

Key insights from the workshop on standards, methodology and implementation for AI in official statistics

What do we actually want to achieve in terms of standards, methodology and implementation for AI in official statistics? This was the main question discussed in a dedicated focus group held in June. Participants put forward their views, priorities and expectations, revealing a wide range of perspectives on where we should be heading next. Quality comes first: it is the guiding principle. Methodological and implementation requirements should follow from these quality requirements.

Feedback under the heading **accuracy and robustness** focused on their measurement, data quality issues and data drift, as well as on how monitoring can be facilitated.

Feedback under the heading **explainability** covered conceptual clarification, the importance of individual words in text data, variable importance, trust and reliability.

Feedback on **reproducibility** included, among other things, consideration of the production pipeline and the skills required. Reproducing an entire pipeline is more complex than reproducing individual steps, particularly within NSIs.

On the topic of **effectiveness**, participants mentioned the acceptance of new approaches, long-term infrastructure, maintenance and support issues, the question of measurability, and innovation through automation.

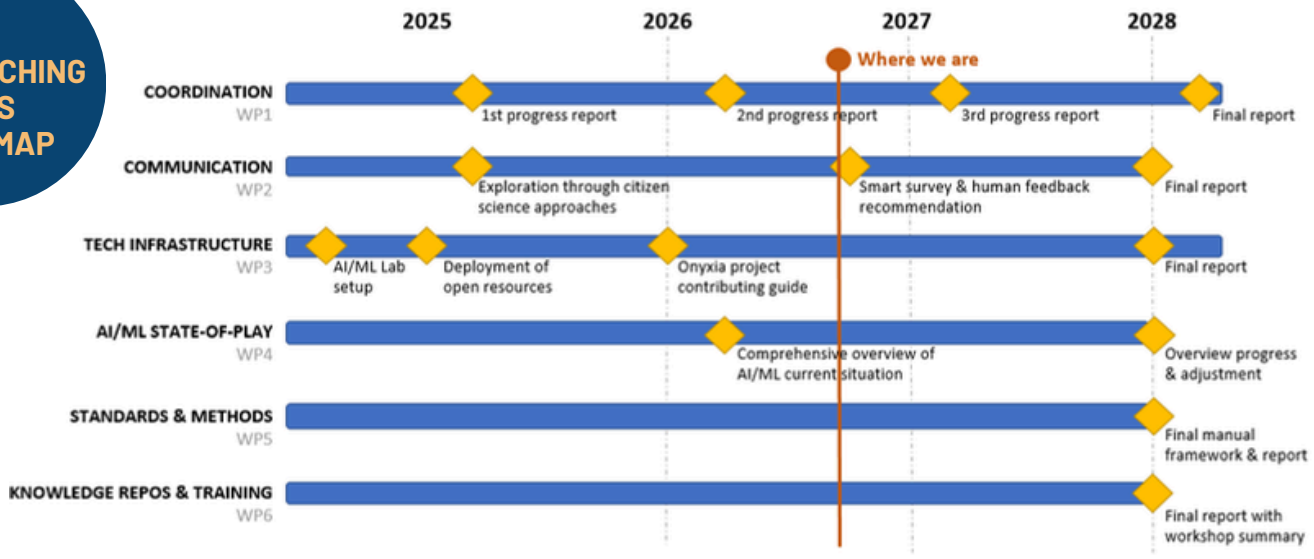


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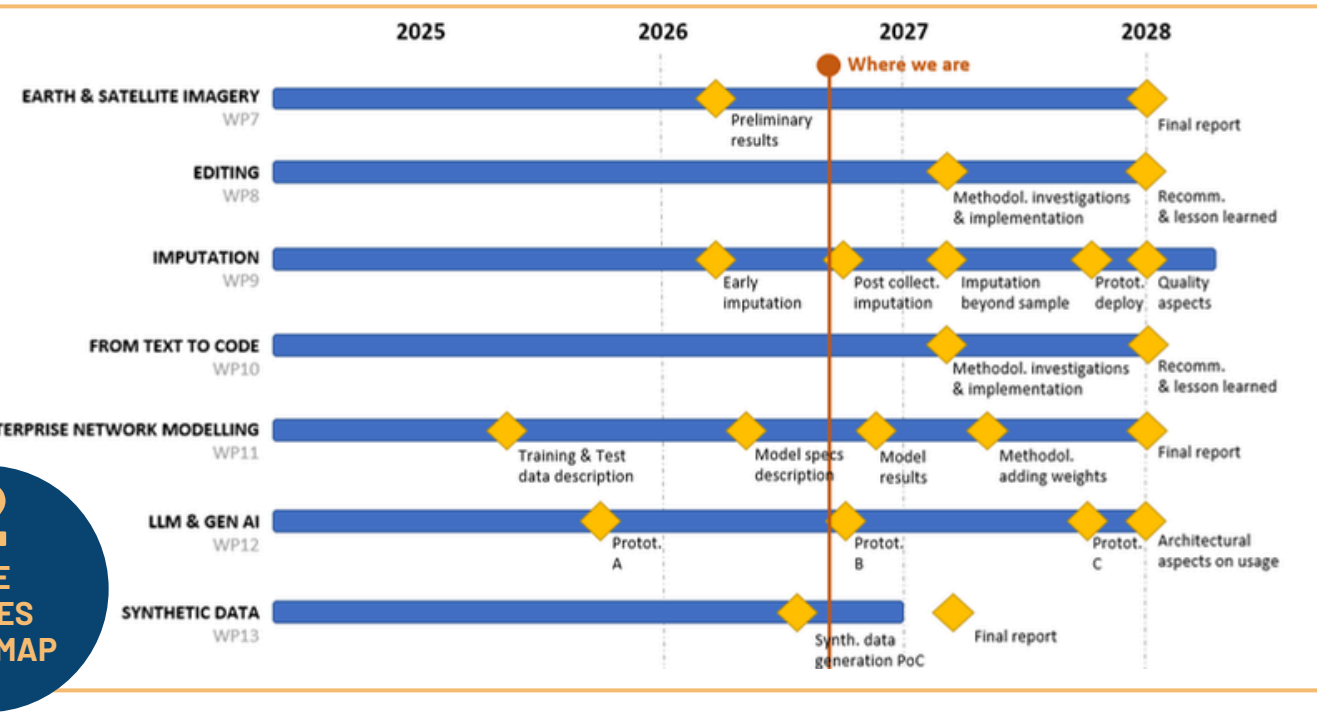
For each WP involved, the project is divided into several phases. Below are descriptions of what will be achieved and when.

PROJECT OVERVIEW

1 OVERARCHING WPS ROADMAP



2 USE CASES ROADMAP



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